MSD MANUAL
Consumer Version

Influenza (Flu)

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Influenza (flu) is a viral infection of the lungs and airways with one of the influenza viruses. It causes a fever, runny nose, sore throat, cough, headache, muscle aches (myalgias), and a general feeling of illness (malaise).

- The virus is spread by inhaling droplets coughed or sneezed out by an infected person or by having direct contact with an infected person's nasal secretions.
- Influenza often starts with chills, followed by a fever, muscle aches, headache, a sore throat, a cough, a runny nose, and a general feeling of illness.
- Influenza is often diagnosed based on symptoms.
- Resting, drinking plenty of fluids, and avoiding exertion can help people recover, as can taking pain relievers, decongestants, and sometimes antiviral medications.
- An annual influenza vaccination is the best way to prevent influenza.

Influenza is distinctly different from the common cold. It is caused by a different virus and produces symptoms that are more severe. Also, influenza affects cells much deeper down in the respiratory tract.

Transmission of influenza

The influenza virus is spread by

- Inhaling droplets that have been coughed or sneezed out by an infected person
- Having direct contact with an infected person's nasal secretions
- Handling household articles that have been in contact with an infected person or an infected person's secretions

Influenza types and strains

There are 3 types of influenza virus:

- Type A
- Type B
- Type C

There are many different strains within types A and B influenza viruses but all cause similar illness. Different strains cause the regular seasonal outbreaks of flu. Type C does not cause a typical influenza illness.

Type A causes most influenza cases (usually more than 70% in a typical season), and most of the others are caused by type B. Type C influenza occurs less frequently, mainly in children.

The strain of influenza virus causing outbreaks is always changing slightly, so that each year the influenza virus is a little different from the previous year's. It often changes enough that previously effective vaccines no longer work.

Influenza type A strains are named based on the specific versions of two proteins present on the surface of the virus. The proteins are H (for hemagglutinin) and N (for neuraminidase). There are 18 different H proteins and 11 N proteins. Thus, a strain might be named influenza A, H1N1. An H1N1 strain was responsible for the so-called swine flu pandemic in 2009–2010. (A pandemic is a major worldwide epidemic.) More recently, H3N2 strains have been causing most infections in the United States.

The name of a strain often reflects its type, the location where it first appeared (for example, Hong Kong flu) or an animal (for example, swine flu), and the year it was detected.

Influenza epidemics and pandemics

In an **influenza epidemic,** many people get sick within a very brief period of time. Every year, throughout the world, widespread outbreaks of influenza occur during late fall or early winter in temperate climates (called seasonal epidemics). Influenza epidemics may occur in two waves:

- First, in schoolchildren and the people who live with them
- Second, in people who are confined to home or who live in long-term care facilities, mainly older adults

In each epidemic, usually only one strain of influenza virus is responsible for the disease.

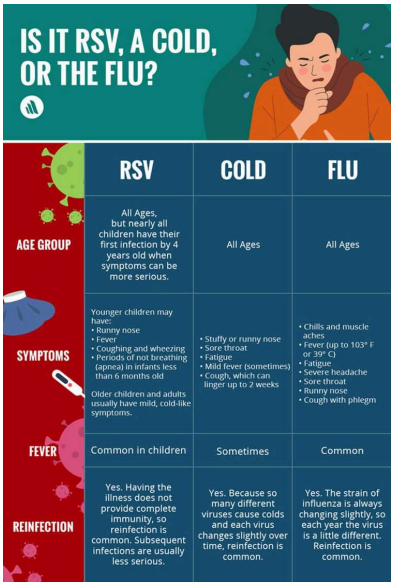
An **influenza pandemic** refers to an outbreak that has spread across a large region, typically across continents and sometimes even worldwide. There have been only 6 major influenza pandemics since 1889. An influenza pandemic is concerning because it usually occurs only when there has been a larger than usual change in the strain of influenza virus. When such a large change occurs, that strain of influenza may affect many more people and cause more severe illness. The risk of death is higher. Although no one knows for certain, scientists think the 1918 influenza pandemic caused 30 to 50 million deaths worldwide, including about 675,000 in the United States.

In 2009–2010, there was an epidemic of the H1N1 strain of influenza virus (see Influenza types and strains) that became widespread and was considered a pandemic. This strain had a combination of genes from pig (swine), bird, and human influenza viruses. Because the first reports focused on the swine component, it was referred to publicly as "swine flu," although it was not acquired directly from pigs. People acquired this influenza virus infection from other infected people (person to person spread) like ordinary flu.

Symptoms of Influenza

Is it RSV, a Cold, or the Flu?

IMAGE



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Symptoms of influenza start 1 to 4 days after infection and can begin suddenly. Chills or a chilly sensation is often the first indication. Fever is common during the first few days, sometimes reaching 102 to 103° F (about 39° C). Many people feel so ill, weak, and tired that they remain in bed for days. They have aches and pains throughout the body, particularly in the back and legs. Headache is often severe, with aching around and behind the eyes. Bright light may make the headache worse.

At first, respiratory symptoms may be relatively mild. They may include a scratchy sore throat, a burning sensation in the chest, a dry cough, and a runny nose. Later, the cough can become severe and bring up phlegm (sputum).

The skin may be warm and flushed, especially on the face. The mouth and throat may redden, the eyes may water, and the whites of the eyes may become bloodshot. People, especially children, may have nausea and vomiting. A few people lose their sense of smell for a few days or weeks. Rarely, the loss is permanent.

Most symptoms subside after 2 or 3 days. However, fever sometimes lasts up to 5 days. Cough, weakness, sweating, and fatigue may persist for several days or occasionally weeks. Mild airway irritation, which can result in a decrease in how long or hard a person can exercise, or slight wheezing may take 6 to 8 weeks to completely resolve.

Complications

The most common complication of influenza is

- Pneumonia

In viral pneumonia, the influenza virus itself spreads into the lungs. Some people with viral pneumonia also become infected with bacteria and develop bacterial pneumonia; this can happen when bacteria (such as pneumococci or staphylococci) attack the person's weakened defenses. With either, people may have a worsened cough, difficulty breathing, persistent or recurring fever, and sometimes blood or pus in the sputum.

People at high risk of complications and death from influenza include

- Children under age 5 years; children under age 2 years are at particularly high risk
- Adults over age 65 years
- People with chronic medical disorders (particularly diabetes mellitus and disorders that affect the heart, lungs, kidneys, liver, or immune system)
- People with severe obesity (body mass index [BMI] of 40 or more)
- Women in the second or third trimester of pregnancy and up to 2 weeks after delivery
- People with disorders that increase their risk of choking on oral secretions, such as stroke or other neurologic disorders that cause weakness, and seizure disorders

Diagnosis of Influenza

- A doctor's evaluation
- Sometimes tests on samples of blood or respiratory secretions
- Sometimes a chest x-ray and measurement of oxygen levels in the blood

Because most people are familiar with the symptoms of influenza and because influenza occurs in epidemics, it is often correctly diagnosed by the person who has it or by family members. The severity of symptoms and the presence of a high fever and body aches help distinguish influenza from a common cold, especially when the illness occurs during an influenza outbreak. It is more difficult to correctly identify influenza by symptoms alone when no outbreak is occurring.

LAB TEST

Flu (Influenza) Test

Tests on samples of respiratory secretions can be used to identify the influenza virus. Blood tests may help determine how sick a person is with the infection. Such tests are done mainly when people appear very ill or when a doctor suspects another cause for the symptoms. Some tests can be done in the doctor's office.

If doctors suspect that pneumonia has developed, they take a chest x-ray and measure oxygen levels in the blood with a sensor placed on a finger (pulse oximetry).

Treatment of Influenza

- Rest and plenty of fluids

The main treatment for influenza is to rest adequately, drink plenty of fluids, and avoid exertion. Normal activities may resume 24 to 48 hours after the body temperature returns to normal, but most people take several more days to recover.

People may treat fever and aches with acetaminophen or nonsteroidal anti-inflammatory drugs (NSAIDs), such as aspirin or ibuprofen. Because of the risk of Reye syndrome, children and adolescents (aged 18 years and under) should not be given aspirin. Acetaminophen and ibuprofen can be used in children if needed. Other measures as listed for the common cold, such as nasal decongestants, may help relieve symptoms.

Antiviral medications (oseltamivir, zanamivir, baloxavir marboxil, and peramivir) are helpful in treating people who have influenza. However, these medications work only if taken within the first day or two after symptoms begin. Antiviral medications reduce severity of symptoms and shorten the duration of fever and the time to return to normal activities, but only by a day or so. Nevertheless, these medications are very effective in some people.

Which medication or medications are used depends on the specific influenza virus causing the infection. Oseltamivir and baloxavir marboxil, taken by mouth, and zanamivir, taken by inhaler, are effective against influenza type A and type B viruses. Oseltamivir may be used in children as young as 1 year old. Zanamivir can be used in adults and children aged 7 years and older, and baloxavir can be used in adults and children aged 12 years and older. Peramivir is given by injection into a vein (intravenously) as a single dose and can be used in adults and children aged 2 years and older who cannot tolerate medications taken by mouth or by inhaler.

If a bacterial infection develops, doctors prescribe antibiotics.

Prevention of Influenza

Prevention consists of

- Vaccination each year for everyone 6 months of age or older (with rare exceptions)
- Sometimes antiviral medications

Prevention is important for all people, but particularly for health care workers and those who are at high risk of complications of influenza.

Did You Know...

- All people 6 months old and over, with rare exceptions, should get a flu vaccination each year.

Vaccines for influenza

Annual vaccination is the best way to avoid getting influenza.

For a detailed discussion of influenza vaccines, see Influenza Vaccine. Vaccination is indicated for most people but is especially important for high-risk people and health care professionals.

Influenza vaccines usually protect against 3 or 4 different strains of influenza virus. The strains of virus that cause influenza outbreaks change each year. Thus, different vaccines are developed each year to keep up with changes in the virus. Experts try to predict the strain of virus that will attack each year based on the strain of virus that predominated during the previous influenza season and the strain causing disease in other parts of the world. When the H and N proteins in the vaccine match those in the influenza strains causing the current epidemic, the vaccine decreases the rate of infection by 70 to 90% in healthy adults.

In older adults who live in long-term care facilities, the vaccine is less likely to prevent influenza, but it reduces the chances of developing pneumonia and of dying. Because the immune system becomes weaker as people age, a high-dose influenza vaccine has been designed specifically for people 65 years and older. This high-dose vaccine can stimulate a stronger immune response in older adults.

Influenza vaccine can be given at the same time as COVID-19 vaccine.

Antiviral medications

Although vaccination is the preferred method of prevention, several antiviral medications can be used in certain people to prevent infection with the influenza virus.

During an outbreak of influenza, antiviral medications are given to people who have been vaccinated within the previous 2 weeks (because it takes 2 weeks for the vaccine to become effective). They are stopped 2 weeks after people are vaccinated. These medications are also given to people who have conditions that make vaccination ineffective or dangerous.

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